

Roméo Maignal

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I am a Computer Science student at EPFL and a Research Assistant at the DATA Lab. I am particularly interested in leveraging functional programming paradigms within Rust to build robust applications such as Embedded Systems, Operating Systems and high-performance CLI tools.

Education

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| ETH Zurich
Master in Computer Science | Zurich, Switzerland
Sept 2026 - ongoing |
| <ul style="list-style-type: none">Relevant coursework: Data-Management Systems, Cloud Computing Architecture, Big Data, Advanced ML/DL, System Security, High-Performance Computing, Advanced Operating Systems, Principles of Distributed Computing | |
| École Polytechnique Fédérale de Lausanne
Bachelor in Computer Science | Lausanne, Switzerland
Sept 2022 - July 2026 |
| <ul style="list-style-type: none">Relevant coursework: Computer Architecture, Computer Systems & Network, Intro to ML, Software Construction, Digital System Design, OOP, Data-Intensive Systems, Computer Language Processing, Parallelism and Concurrency | |

Work Experience

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| EPFL DATA Laboratory ↗
Research Assistant | Lausanne, Switzerland
Febr 2026 - June 2026 |
| <ul style="list-style-type: none">Implemented the interpretation of SPARQL for the superfluid database management system.Wrote a Parser that transformed queries into an interpretable Abstract Syntax Tree that I then transformed to its Datalog equivalent.Optimized SPARQL-specific predicates throughout the DBMS pipeline. | |
| EPFL Spacecraft Team ↗
System Engineer for the CHESS Mission | Lausanne, Switzerland
Sept 2023 - Febr 2025 |
| <ul style="list-style-type: none">Contributed to the design review by writing Interface Control Documents for 4 different subsystems following ECSS standards.Successfully troubleshooted and resolved technical issues that arose during the satellite's design review such as misplacement of crucial mechanical interfaces. | |

Notable Projects

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| Aircraft Flights Tracking Software ↗ |
| <ul style="list-style-type: none">Developed a FlightRadar24-like software from scratch in Java over a period of four months. It involved various engineering fields (data transmission, aircraft data interfaces, ADS-B communication, aircraft location settings, cryptography).Tools used: Java, JavaFX, JUnit, JVM, IntelliJ, ADS-B Antenna |
| Autonomous 6-legged walking robot ↗ |
| <ul style="list-style-type: none">Built a small robot capable of locating and tracking a person within a room, using both sound recognition and computer vision.Tools used: Raspberry Pi, Esp32, Python, OnShape, Prusa-Slicer |
| Scala Web App ↗ |
| <ul style="list-style-type: none">Multiplayer web game application. It implements a famous board game and uses a client-server architecture to connect multiple players. Made use of Scala state machines and json serialization/deserialization functions.Tools used: Scala, Metals, SBT |
| Interface Control Documents ↗ |
| <ul style="list-style-type: none">ICDs for the telecommunication subsystems of a CubeSat. It involved learning the specific requirements for CubeSat development and adhering to ECSS standards to give the project good technical directives.Tools used: Overleaf, LaTeX, FusionCAD |
| Roguelike Java Game ↗ |
| <ul style="list-style-type: none">Roguelike desktop game built in Java.Tools used: Java, JavaFX, JUnit, IntelliJ |

Competences

Programming Languages: Rust, Scala, Java, Python, TypeScript/React, C, CUDA C++, Go, RISC-V Assembly, Verilog, LaTeX

Tools: Git, Github, Vim, Linux, VSCode, Overleaf, GTKWave, OnShape, Blender

Skills: Functional Programming, Systems/Network Programming, Low-Level Programming, High Performance Parallel Programming, Algorithms, Data Structures, CI/CD, Web Development

Languages: French (Native), English (C1 - IELTS), Spanish (B1)